

CODE LAYOUT, READABILITY AND REUSABILITY

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

Rising Waters is developed by following coding standards that improve readability, maintainability, and code reusability. The project is organized into separate modules, making it easy to understand, test, and enhance in the future.

Code Layout

The project follows a structured folder organization:

- app.py – Main Streamlit application
- model.pkl – Trained Machine Learning model
- requirements.txt – Project dependencies
- README.md – Project documentation
- Dataset – Rainfall and weather data
- GitHub Repository – Source code management

Readability

The source code is written using meaningful variable names, proper indentation, comments, and modular functions. These practices make the code easier to understand for developers and reviewers.

Readability Features

- Meaningful variable names
- Proper indentation
- Modular functions
- Comments for important logic
- Consistent coding style

Code Reusability

Reusable functions are created for data preprocessing, prediction, and model loading. This reduces duplicate code and improves maintainability.

Advantages

- Easy to understand
- Easy to maintain
- Faster debugging
- Better collaboration

- Supports future enhancements

Best Practices Followed

- Python coding standards
- Modular programming
- Error handling
- Version control using GitHub
- Clean project structure

Conclusion

The Rising Waters project follows good software engineering practices by maintaining a clean code layout, improving readability, and encouraging code reuse. These practices make the application scalable, maintainable, and suitable for future development